



Apache Kafka Installation and Validation Guide

Overview

This document provides a **complete step-by-step guide** to installing and validating **Apache Kafka** on a local system. Two supported installation methods are covered:

1. **Local installation using Kafka binaries**
2. **Docker-based installation using Docker Compose**

After completing this guide, you will be able to:

- Run Kafka locally or using Docker
- Create Kafka topics
- Produce and consume messages
- Verify a working Kafka setup

Prerequisites

Before starting, ensure the following prerequisites are met.

Common Requirements

- Stable internet connection
- Basic familiarity with command-line tools (Terminal, Command Prompt, or PowerShell)

For Local Installation

- **Java 8 or Java 11** installed
- `java` added to system PATH

For Docker Installation

- Docker Desktop installed
- Docker daemon running



- Docker Compose enabled

Note: Installation steps may vary based on operating system (Windows, macOS, Linux), Java version, Docker version, system permissions, and network restrictions.

Choosing an Installation Method

Method	Recommended When
Local (Kafka binaries)	You want to understand Kafka internals and configuration
Docker (Docker Compose)	You want a quick, isolated, production-like setup

Method 1: Local Installation (Kafka Binary Distribution)

Step 1: Verify Java Installation

Kafka requires Java to run.

Check the Java version:

```
java -version
```

Ensure the output shows **Java 8 or Java 11**.

Step 2: Download Apache Kafka

1. Visit the official Apache Kafka download page.
2. Download the **latest stable Kafka version**.
 - a. The package name includes a Scala version (for example, kafka_2.13-<version>).
3. Choose:
 - a. **ZIP file** for Windows



- b. **TGZ file** for macOS/Linux

Step 3: Extract Kafka

- **Windows:** Extract the ZIP file using File Explorer.
- **macOS / Linux:**

```
tar -xvf kafka_<version>.tgz
```

Step 4: Navigate to the Kafka Directory

```
cd kafka_<version>
```

This directory contains all Kafka binaries and configuration files.

Step 5: Start Kafka Broker

Open a new terminal and run:

```
bin/kafka-server-start.sh config/server.properties
```

On Windows, use:

```
bin/windows/kafka-server-start.bat config/server.properties
```

Kafka will start and listen on **port 9092** by default.

Step 6: Verify Kafka Is Running

If Kafka starts successfully:

- No errors appear in the terminal
- The broker remains running



- Port **9092** is active

Kafka is now ready for topic creation and message exchange.

Method 2: Docker Installation (Docker Compose)

Step 1: Verify Docker Installation

Run the following commands:

```
docker --version  
docker compose version
```

Ensure Docker is running before proceeding.

Step 2: Create a Project Directory

```
mkdir kafka-docker  
cd kafka-docker
```

Step 3: Create Docker Compose File

Create a file named `docker-compose.yml` and define services for:

- Kafka
- Required dependencies (such as Zookeeper or Kraft-based setup)

Ensure:

- Port **9092** is exposed
- Kafka is accessible from the host machine



Step 4: Start Kafka Using Docker

Run:

```
docker compose up -d
```

This command:

- Pulls required Docker images
- Starts Kafka containers
- Exposes Kafka broker on port 9092

Step 5: Verify Running Containers

Check container status:

```
docker ps
```

Ensure Kafka containers are running and not restarting or exiting.

Topic Creation and Validation

Once Kafka is running (local or Docker), validate the setup by creating a topic and producing/consuming messages.

Step 1: Create a Kafka Topic

Local Installation

```
bin/kafka-topics.sh --create </span>  
--topic demo-topic </span>  
--bootstrap-server localhost:9092 </span>  
--partitions 1 </span>
```



```
--replication-factor 1
```

Docker Installation

```
docker exec -it <kafka_container_name> kafka-topics </span>  
--create </span>  
--topic demo-topic </span>  
--bootstrap-server localhost:9092 </span>  
--partitions 1 </span>  
--replication-factor 1
```

Step 2: Produce Messages

Start a producer:

Local Installation

```
bin/kafka-console-producer.sh </span>  
--topic demo-topic </span>  
--bootstrap-server localhost:9092
```

Type a few sample messages and press Enter after each.

Docker Installation

```
docker exec -it <kafka_container_name> kafka-console-producer </span>  
--topic demo-topic </span>  
--bootstrap-server localhost:9092
```

Step 3: Consume Messages

Open a new terminal and run:



Local Installation

```
bin/kafka-console-consumer.sh </span>  
--topic demo-topic </span>  
--bootstrap-server localhost:9092 </span>  
--from-beginning
```

Docker Installation

```
docker exec -it <kafka_container_name> kafka-console-consumer </span>  
--topic demo-topic </span>  
--bootstrap-server localhost:9092 </span>  
--from-beginning
```

If messages appear, Kafka is functioning correctly.

Validation Checklist

Kafka broker is running
Topic created successfully
Messages produced without errors
Messages consumed successfully

Completion of these steps confirms a valid Kafka installation.

Stopping Kafka

Local Installation

- Press Ctrl + C in the Kafka broker terminal



Docker Installation

docker compose down

Common Troubleshooting

Issue	Possible Cause
java not recognised	Java not added to PATH
Port 9092 already in use	Another service using the port
Broker connection refused	Kafka not running
Docker containers exit	Docker daemon not running
Unable to create topics	Kafka broker not reachable

Next Steps

After successful installation and validation, you are ready to:

- Explore Kafka configuration options
- Work with producers and consumers programmatically
- Build real-time streaming pipelines
- Integrate Kafka with Spark, Flink, or other systems

This setup forms the foundation for all Kafka-based streaming applications.

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